

- (b) Refine the previous analysis by showing that for every integer $N > 0$ one has the following asymptotic expansion

$$\text{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + 2\frac{x}{(\log x)^3} \cdots + (N-1)! \frac{x}{(\log x)^N} + O\left(\frac{x}{(\log x)^{N+1}}\right)$$

as $x \rightarrow \infty$.

11. Let

$$\varphi(x) = \sum_{p \leq x} \log p$$

where the sum is taken over all primes $\leq x$. Prove that the following are equivalent as $x \rightarrow \infty$:

- (i) $\varphi(x) \sim x$,
- (ii) $\pi(x) \sim x/\log x$,
- (iii) $\psi(x) \sim x$,
- (iv) $\psi_1(x) \sim x^2/2$.

12. If p_n denotes the n^{th} prime, the prime number theorem implies that $p_n \sim n \log n$ as $n \rightarrow \infty$.

- (a) Show that $\pi(x) \sim x/\log x$ implies that

$$\log \pi(x) + \log \log x \sim \log x.$$

- (b) As a consequence, prove that $\log \pi(x) \sim \log x$, and take $x = p_n$ to conclude the proof.

4 Problems

1. Let $F(s) = \sum_{n=1}^{\infty} a_n/n^s$, where $|a_n| \leq M$ for all n .

- (a) Then

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |F(\sigma + it)|^2 dt = \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{2\sigma}} \quad \text{if } \sigma > 1.$$

How is this reminiscent of the Parseval-Plancherel theorem? See e.g. Chapter 3 in Book I.

- (b) Show as a consequence the uniqueness of Dirichlet series: If $F(s) = \sum_{n=1}^{\infty} a_n n^{-s}$, where the coefficients are assumed to satisfy $|a_n| \leq cn^k$ for some k , and $F(s) \equiv 0$, then $a_n = 0$ for all n .

Hint: For part (a) use the fact that

$$\frac{1}{2T} \int_{-T}^T (nm)^{-\sigma} n^{-it} m^{it} dt \rightarrow \begin{cases} n^{-2\sigma} & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$$

2.* One of the “explicit formulas” in the theory of primes is as follows: if ψ_1 is the integrated Tchebychev function considered in Section 2, then

$$\psi_1(x) = \frac{x^2}{2} - \sum_{\rho} \frac{x^{\rho}}{\rho(\rho+1)} - E(x)$$

where the sum is taken over all zeros ρ of the zeta function in the critical strip. The error term is given by $E(x) = c_1 x + c_0 + \sum_{k=1}^{\infty} x^{1-2k} / (2k(2k-1))$, where $c_1 = \zeta'(0)/\zeta(0)$ and $c_0 = \zeta'(-1)/\zeta(-1)$. Note that $\sum_{\rho} 1/|\rho|^{1+\epsilon} < \infty$ for every $\epsilon > 0$, because $(1-s)\zeta(s)$ has order of growth 1. (See Exercise 8.) Also, obviously $E(x) = O(x)$ as $x \rightarrow \infty$.

3.* Using the previous problem one can show that

$$\pi(x) - \text{Li}(x) = O(x^{\alpha+\epsilon}) \quad \text{as } x \rightarrow \infty$$

for every $\epsilon > 0$, where α is fixed and $1/2 \leq \alpha < 1$ if and only if $\zeta(s)$ has no zeros in the strip $\alpha < \text{Re}(s) < 1$. The case $\alpha = 1/2$ corresponds to the Riemann hypothesis.

4.* One can combine ideas from the prime number theorem with the proof of Dirichlet’s theorem about primes in arithmetic progression (given in Book I) to prove the following. Let q and ℓ be relatively prime integers. We consider the primes belonging to the arithmetic progression $\{qk + \ell\}_{k=1}^{\infty}$, and let $\pi_{q,\ell}(x)$ denote the number of such primes $\leq x$. Then one has

$$\pi_{q,\ell}(x) \sim \frac{x}{\varphi(q) \log x} \quad \text{as } x \rightarrow \infty,$$

where $\varphi(q)$ denotes the number of positive integers less than q and relatively prime to q .

8 Conformal Mappings

The results I found for polygons can be extended under very general assumptions. I have undertaken this research because it is a step towards a deeper understanding of the mapping problem, for which not much has happened since Riemann's inaugural dissertation; this, even though the theory of mappings, with its close connection with the fundamental theorems of Riemann's function theory, deserves in the highest degree to be developed further.

E. B. Christoffel, 1870

The problems and ideas we present in this chapter are more geometric in nature than the ones we have seen so far. In fact, here we will be primarily interested in mapping properties of holomorphic functions. In particular, most of our results will be “global,” as opposed to the more “local” analytical results proved in the first three chapters. The motivation behind much of our presentation lies in the following simple question:

Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?

By a holomorphic bijection we simply mean a function that is both holomorphic and bijective. (It will turn out that the inverse map is then automatically holomorphic.) A solution to this problem would permit a transfer of questions about analytic functions from one open set with little geometric structure to another with possibly more useful properties. The prime example consists in taking $V = \mathbb{D}$ the unit disc, where many ideas have been developed to study analytic functions.¹ In fact, since the disc seems to be the most fruitful choice for V we are led to a variant of the above question:

Given an open subset Ω of $\mathbb{C}$, what conditions on Ω guarantee that there exists a holomorphic bijection from Ω to $\mathbb{D}$?

¹For the corresponding problem when $V = \mathbb{C}$, the solution is trivial: only $U = \mathbb{C}$ is possible. See Exercise 14 in Chapter 3.